

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	HDN PRAHARSHINI
Register Number	192424085
Date	3 october 2026
Day	Thursday

SET 7 – CUSTOMER DEMOGRAPHICS ANALYSIS

Aim

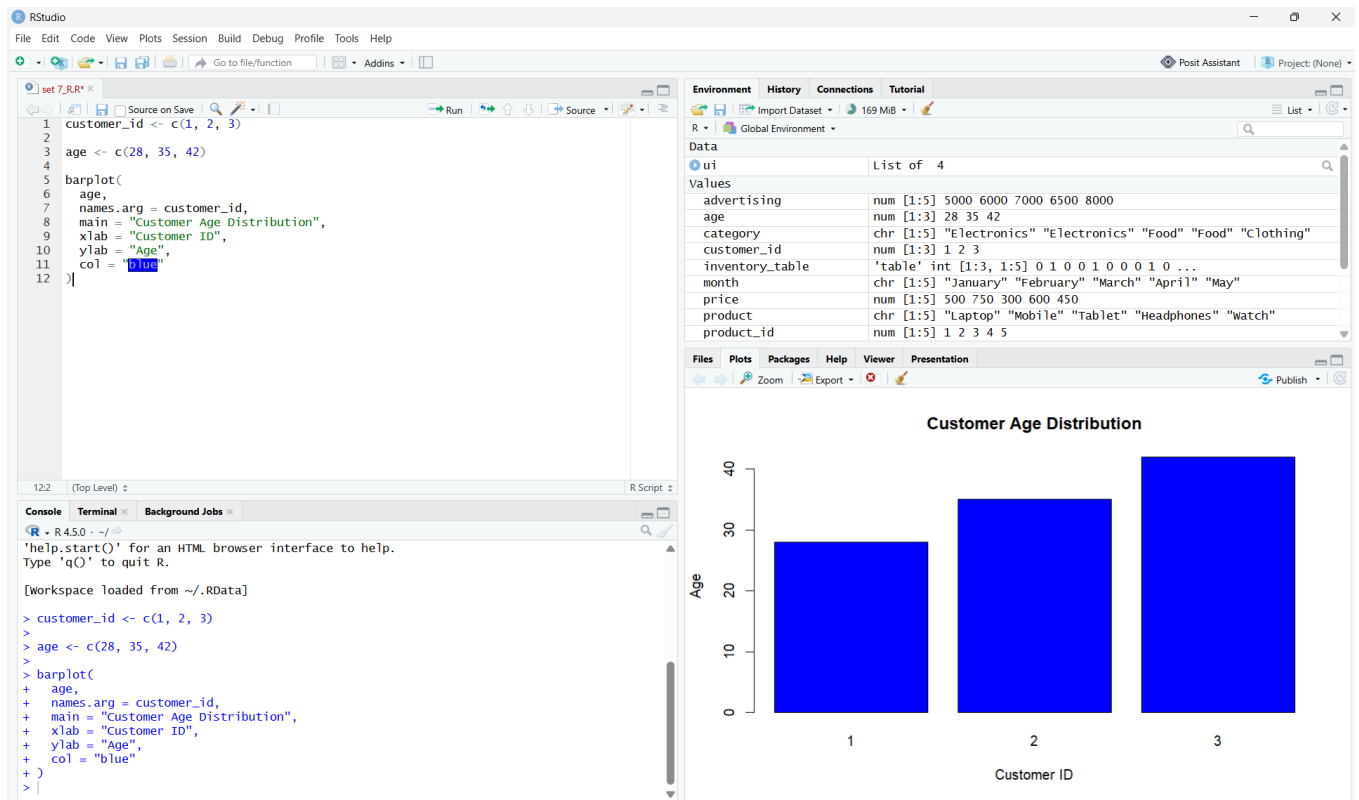
To analyze and visualize customer demographic information using bar charts, pie charts, tables, and an interactive dashboard.

PROGRAM 1: Bar Chart – Distribution of Customer Ages

To create a line chart to visualize the daily page views of the website over time..

```
customer_id <- c(1, 2, 3)

age <- c(28, 35, 42)
barplot(
  age,
  names.arg = customer_id,
  main = "Customer Age Distribution",
  xlab = "Customer ID",
  ylab = "Age",
  col = "blue"
)
```



Output 1: Daily website page views

Observation: The bar chart shows the ages of the three customers. Customer 3 has the highest age of **42 years**, while Customer 1 has the lowest age of **28 years**.

PROGRAM 2: Pie Chart – Gender Distribution

To create a pie chart to represent the distribution of customers based on gender.

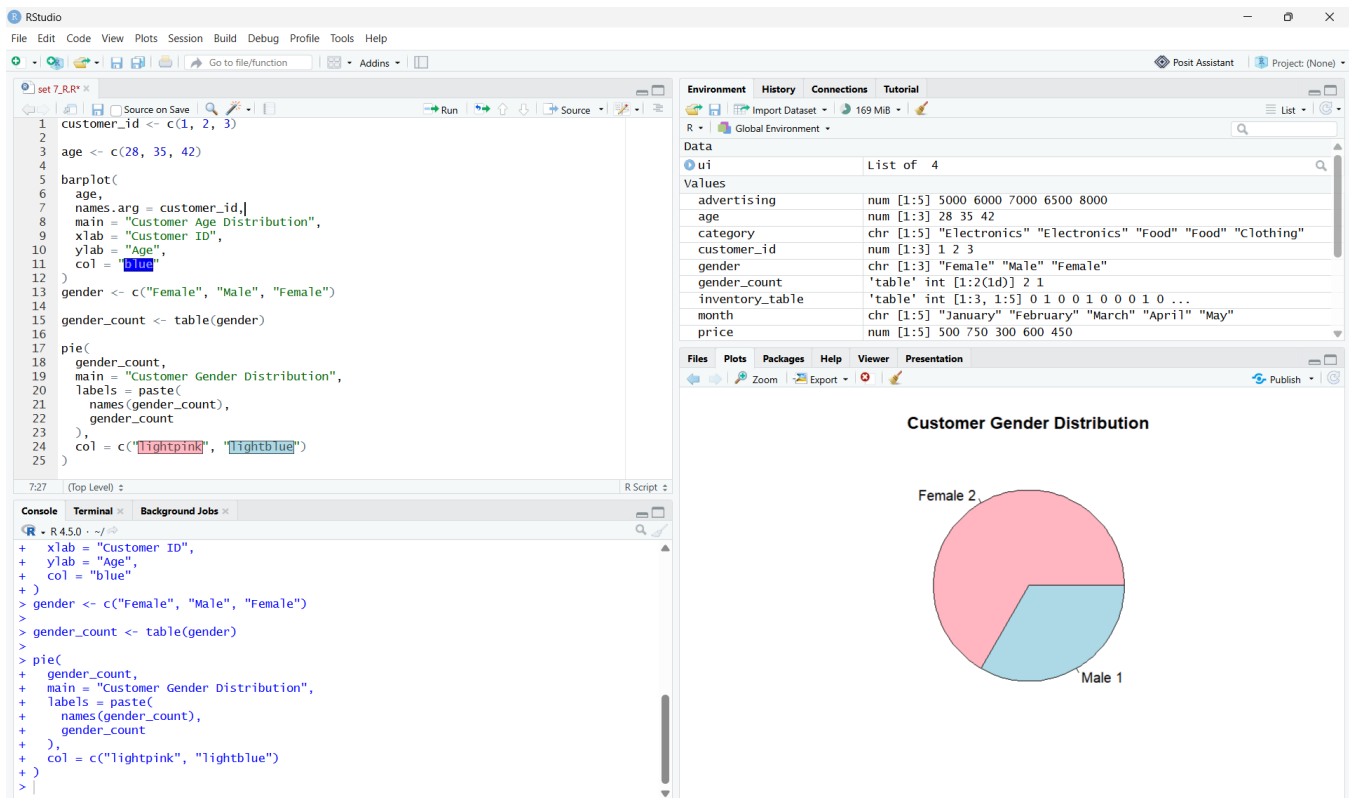
```

gender <- c("Female", "Male", "Female")

gender_count <- table(gender)

pie(
  gender_count,
  main = "Customer Gender Distribution",
  labels = paste(
    names(gender_count),
    gender_count
  ),
  col = c("lightpink", "lightblue")
)

```



Output 2: click through rate by day

Observation: There are **2 female customers** and **1 male customer** in the given dataset. Therefore, females represent the larger portion of the pie chart.

PROGRAM 3: Table – Customer Demographic Information

To create a table displaying the demographic information of each customer..

```

customer_id <- c(1, 2, 3)
age <- c(28, 35, 42)
gender <- c(
  "Female",
  "Male",
  "Female"
)
income <- c(
  50000,
  60000,
  75000
)
customer_table <- data.frame(
  Customer_ID = customer_id,

```

```

Age = age,
Gender = gender,
Income = income
)

```

The screenshot shows the RStudio interface. The source editor contains R code that creates a data frame with three rows of customer data. The console shows the execution of this code, resulting in a printed table of the data.

```

set 7_R.R* x
46:19 (Top Level) R Script

25 )
26
27 customer_id <- c(1, 2, 3)
28
29 age <- c(28, 35, 42)
30
31 gender <- c(
32   "Female",
33   "Male",
34   "Female"
35 )
36
37 income <- c(
38   50000,
39   60000,
40   75000
41 )
42
43 customer_table <- data.frame(
44   Customer_ID = customer_id,
45   Age = age,
46   Gender = gender,
47   Income = income
48 )
49
50 customer_table

```

```

+ 50000,
+ 60000,
+ 75000
+ )
>
> customer_table <- data.frame(
+   Customer_ID = customer_id,
+   Age = age,
+   Gender = gender,
+   Income = income
+ )
>
> customer_table
  Customer_ID Age Gender Income
1           1  28 Female 50000
2           2  35  Male 60000
3           3  42 Female 75000
>

```

Output 3: user interactions

Observation: The table successfully displays the Customer ID, age, gender, and income of all three customers.

PROGRAM 4: Interactive Dashboard

To develop an interactive dashboard combining the customer age bar chart, gender pie chart, and demographic table.

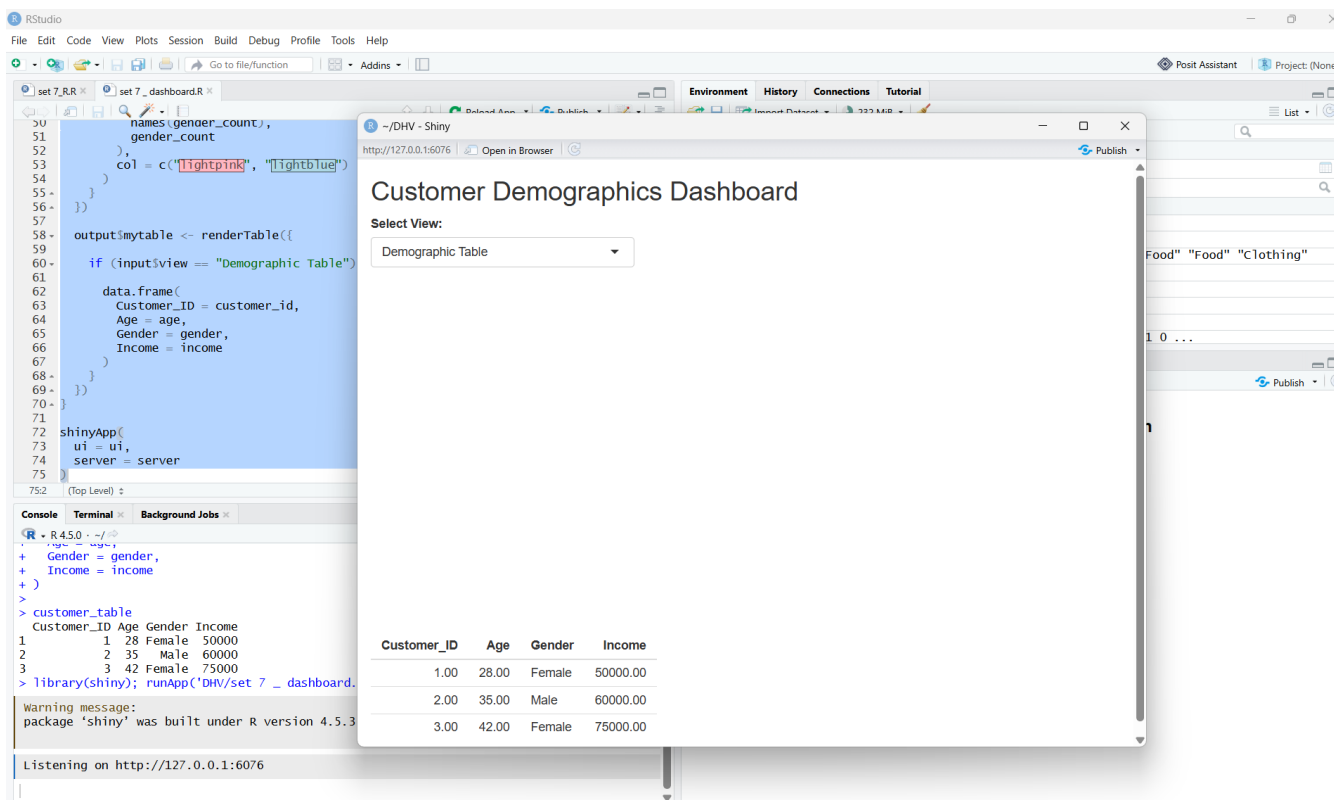
```
library(shiny)
customer_id <- c(1, 2, 3)
age <- c(28, 35, 42)
gender <- c("Female", "Male", "Female")
income <- c(50000, 60000, 75000)
ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),
  selectInput(
    "view",
    "Select View:",
    choices = c(
      "Age Distribution",
      "Gender Distribution",
      "Demographic Table"
    )
  ),
  plotOutput("myplot"),
  tableOutput("mytable")
)
server <- function(input, output) {
  output$myplot <- renderPlot({
    if (input$view == "Age Distribution") {
      barplot(
        age,
        names.arg = customer_id,
        main = "Customer Age Distribution",
        xlab = "Customer ID",
        ylab = "Age",
        col = "blue"
      )
    }

    if (input$view == "Gender Distribution") {
```

```

gender_count <- table(gender)
pie(
  gender_count,
  main = "Customer Gender Distribution",
  labels = paste(
    names(gender_count),
    gender_count
  ),
  col = c("lightpink", "lightblue")
)
}
}
output$mytable <- renderTable(
  if (input$view == "Demographic Table") {
    data.frame(
      Customer_ID = customer_id,
      Age = age,
      Gender = gender,
      Income = income
    )
  }
})
}
shinyApp(
  ui = ui,
  server = server
)
)

```



Output 4: website analytics dashboard

Observation: The dashboard allows the user to interactively view customer age, gender distribution, and demographic information.

Set 7 Procedure

- Create vectors for customer ID, age, gender, and income.
- Use `barplot()` to visualize customer ages.
- Use `table()` and `pie()` to represent gender distribution.
- Use `data.frame()` to create a demographic information table.
- Use Shiny to combine the charts and table into an interactive dashboard.
- Save each question as a separate R script and run the programs in RStudio.
-

Set 7 Result

Customer demographic data was successfully analyzed and visualized using a bar chart, pie chart, and table. An interactive dashboard was also developed to explore customer demographic information.